

RESEARCH STATEMENT

Willie Rush Lim

My research is in one-dimensional complex dynamics, which lies at the intersection of complex analysis and geometry. I study the properties of dynamical sets associated to holomorphic maps using techniques from renormalization theory. The main emphasis of my work is the near-neutral regime, which is at the forefront of current developments in the field.

Core problems. Complex dynamics is concerned with iterates of holomorphic maps on complex manifolds, and one overarching goal is to understand stability and bifurcation in the space of such maps. In dimension one, the classical study of rational endomorphisms of the Riemann sphere was initiated by Fatou and Julia. A rational map is said to be hyperbolic (Smale’s Axiom A) if its global behavior is dominated by attracting periodic dynamics; such maps are well-understood and structurally stable. The Density of Hyperbolicity Conjecture (DoH) states that hyperbolic maps are dense in the space of rational maps.

For the family of quadratic polynomials $z \mapsto z^2 + c$, the MLC Conjecture on the local connectivity of the Mandelbrot set implies DoH and provides a complete topological description of the Mandelbrot set through its pinched-disk model. MLC is equivalent to the Combinatorial Rigidity Conjecture: roughly speaking, a non-hyperbolic quadratic polynomial should be uniquely determined by its combinatorial data. This problem shares deep roots with the celebrated rigidity theorem of Mostow as well as the ending lamination theorem in hyperbolic geometry.

MLC has further geometric and probabilistic counterparts — such as the conjecture that the boundary of the Mandelbrot set has zero Lebesgue measure — and guides other deep problems in complex dynamics. The main framework for studying these problems is renormalization theory, and, as I discuss below, all of them currently confront the same underlying challenge: near-neutral dynamics.

Renormalization. The basic idea is to rescale a suitable first return map and obtain a new dynamical system of the same type. Iterating this procedure often reveals self-similarity and can convert geometric questions about families of dynamical systems into questions about the dynamics of an operator on a space of maps. In the 1970’s, Feigenbaum, and independently Coullet and Tresser, used this framework to explain the universal scaling laws of period-doubling cascades, suggesting that such universality arises from hyperbolicity of the renormalization operator. This discovery led to Feigenbaum receiving the 1986 Wolf Prize in Physics. Subsequent work of Sullivan, McMullen, Lyubich, and others developed this picture rigorously. With time, it became apparent that universality is just one of the many manifestations of renormalization. For instance, a major culmination of real renormalization theory in the early 2000’s is Lyubich’s theorem that almost every real quadratic polynomial is either *regular* or *stochastic*.

Contributions. One of the principal challenges to resolving these problems for complex rational maps is near-neutral dynamics where the behavior depends with extreme sensitivity on the arithmetic of the rotation number. My research contributes to this problem in several directions. In my PhD thesis and subsequent work, I proved a realization theorem for Herman curves and established rigidity and hyperbolicity results for critical quasicircle maps. A centerpiece of my postdoctoral work is an MLC-type rigidity theorem for the full neutral renormalization attractor for quadratic polynomials, proved jointly with Dudko and Lyubich. These are connected by common themes: geometric bounds, rigidity, and the extension of real renormalization theory to genuinely complex settings.

1. THE RENORMALIZATION THEORY OF HERMAN CURVES

My initial research direction grew out of my PhD thesis where I carried out the program to study the geometry of Herman rings, their degeneration to Herman curves, and the corresponding renormalization theory.

1.1. A priori bounds and degeneration. A *Herman ring* is a periodic annular rotation domain of a rational map. The initial study of Herman rings was tied to KAM theory and the small divisor problem in celestial mechanics. Unlike a *Siegel disk* (a simply connected rotation domain), a Herman ring has a non-trivial deformation space and its geometry is substantially harder to control.

In my Inventiones paper, I obtained *a priori bounds* for Herman rings in the bounded type setting.

Theorem A ([Lim25]). *Under a simple configuration hypothesis, the boundaries of a Herman ring of a rational map with a bounded type irrational rotation number are quasicircles with dilatation depending only on the degree and the rotation number.*

I proved this theorem using near-degenerate conformal-geometric machinery developed by Kahn, Lyubich, and Dudko [KL05, Kah06, DL22]. It leads to the first general construction of non-trivial Herman curves, answering a question posed by Eremenko [Ere20]. A *Herman curve* is an invariant Jordan curve on which the dynamics is conjugate to an irrational rotation but which do not arise as part of the closure of a rotation domain.

Theorem B ([Lim25]). *One can construct rational maps admitting Herman curves with any prescribed bounded type combinatorial data.*

Here, the combinatorial data consist of the bounded type irrational rotation number and the combinatorial location of inner and outer critical points along the Herman curve.

1.2. Rigidity of rational maps. One basic question is how much room a rational map's dynamics leaves for deformation. A rational map admits an *invariant line field* on its Julia set if there is an invariant measurable Beltrami differential supported on a positive measure subset of the Julia set. The absence of invariant line fields rules out any exotic deformation coming from the Julia set. McMullen and Sullivan [McM94, MS98] conjectured that, aside from classical exceptions, every rational map does not admit any invariant line field. Particularly, it implies the DoH conjecture. In [Lim26a], I made a contribution to this problem.

Theorem C ([Lim26a]). *If a rational map is geometrically finite away from finitely many bounded type rotation domains and Herman curves, then it has no invariant line field on its Julia set. In the absence of Herman curves, its Julia set has zero Lebesgue measure.*

This theorem provides a broad extension of earlier rigidity results for geometrically finite maps [Sul83, McM00]. It gives applications to the combinatorial rigidity for the rational maps constructed in Theorem A and for bounded type non-renormalizable Siegel polynomials [DY25].

1.3. Critical quasicircle maps and universality. Theorem C establishes Herman curves as rigid objects for rational maps. The natural next question is to explore beyond the realm of rational maps.

A *(uni-)critical quasicircle map* $f : X \rightarrow X$ is a holomorphic dynamical system on an invariant quasicircle X with a single critical point. Such systems form a complex counterpart of the well-developed theory of real-analytic critical circle maps [Lan88, dF99, dFdM99, Yam01, Yam02, Yam03] where Feigenbaum-type universality was proven rigorously. My work extends this from round circles to quasicircles and allows imbalance at the critical point — a genuinely complex phenomenon with no real analog. Theorem B constructs such maps for any bounded type combinatorial data: rotation number θ , inner criticality $d_0 \geq 2$, and outer criticality $d_\infty \geq 2$. See Figure 1 for an example. I proved that such data determines the map up to some smooth equivalence.

Theorem D ([Lim26a]). *Any two critical quasicircle maps with the same bounded type combinatorics are quasiconformally conjugate on a neighborhood of their invariant quasicircles. Moreover, the conjugacy is uniformly $C^{1+\alpha}$ -conformal along the quasicircle.*

In the proof, I studied the underlying renormalization theory and developed complex *butterfly bounds* for renormalizations. It implies universality of the Hausdorff dimension, self-similarity for quadratic irrational rotation numbers, and exponential convergence of renormalizations.

Rigidity does not address what happens for perturbations of critical quasicircle maps. I explored this problem by proving a Feigenbaum-type hyperbolicity result for these maps.

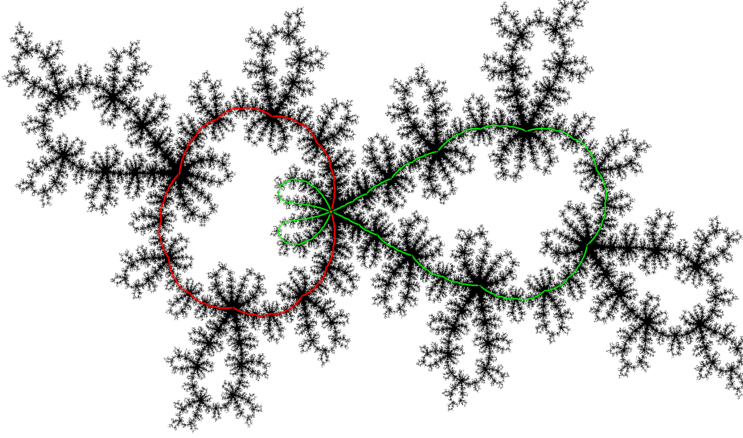


FIGURE 1. The Julia set of a degree 4 rational map that is a critical quasicycle map with golden mean rotation number and imbalanced criticalities $(d_0, d_\infty) = (3, 2)$. The Herman curve is colored red and its strict preimage is colored green.

Theorem E ([Lim26b]). *For every periodic type combinatorics, the corresponding renormalization operator is analytic and admits a hyperbolic fixed point with exactly one unstable direction. Its local stable manifold is the local conjugacy class of the fixed point.*

As a concrete application, critical quasicycle maps with fixed pre-periodic type combinatorics form a codimension at most 1 analytic submanifold of their natural Banach space. The hardest step in the proof is pinning down the exact dimension of the unstable manifold. To address this, I proved a rigidity result on the transcendental structure of maps on the unstable manifold; an analog of this for transcendental entire functions was proven by Rempe [Rem09].

Together, these theorems give a rigidity and universality picture for unicritical Herman curves and suggest a broader program: which features of real renormalization theory survive in the genuinely complex setting, and which change.

2. NEUTRAL RENORMALIZATION

My current research focuses on a second, closely related setting: quadratic polynomials with a neutral fixed point,

$$f_\theta(z) := e^{2\pi i\theta} z + z^2.$$

Depending on the arithmetic properties of θ , the neutral fixed point at 0 may be linearizable or non-linearizable, and the associated dynamical sets can have very different geometric properties. My main achievement to date in this program is an MLC-type rigidity theorem for neutral renormalization across all irrational rotation numbers.

2.1. Sector renormalization. A key development in this direction has been the theory of sector renormalization. For a neutral holomorphic germ $g(z) = e^{2\pi i\theta} z + O(z^2)$, one can construct a fundamental sector near the fixed point at 0 and glue its sides using the dynamics; see Figure 2. The first return map then produces a new neutral holomorphic germ $\mathcal{R}_{\text{sec}}g$, and the procedure can be iterated. Yoccoz [Yoc95] carefully studied this procedure to prove his seminal result on the optimality of the Brjuno condition for linearizability.

More recently, Dudko and Lyubich [DL22, DL26a] applied the near-degenerate regime to establish uniform *a priori* bounds for this renormalization scheme for neutral quadratic polynomials f_θ for all irrational θ . Notably, this revolves around the construction of uniformly quasiconformal *pseudo-Siegel disks* $\hat{Z}_\theta$ that contain both the neutral fixed point and the postcritical set (the closure of the critical

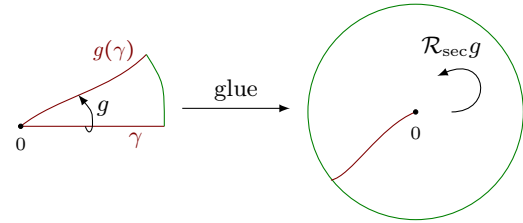


FIGURE 2. Sector renormalization

orbit of f_θ). One main application is the existence of the *Mother Hedgehog* H_θ , a completely f_θ -invariant full planar continuum that contains the neutral fixed point and the critical orbit of f_θ . In [Lim26d], I used pseudo-Siegel disks to study typical orbits of f_θ .

Theorem F ([Lim26d]). *For every $\theta \in \mathbb{R}/\mathbb{Z}$, the postcritical set of f_θ has zero Lebesgue measure.*

Previously, zero area results were only known under various arithmetic restrictions on θ [Dou87, PZ04, Che13, Che19]; my theorem works uniformly across all rotation numbers. The proof combines the pseudo-Siegel bounds, uniform distortion estimates for the sector-gluing maps, and a porosity argument. A direct corollary is that almost every point in the Julia set of f_θ is non-recurrent.

2.2. The full renormalization attractor. Iterated sector renormalizations form an attractor consisting of bi-infinite renormalization towers of neutral holomorphic maps. Controlling this attractor requires a second ingredient: uniform control of the dynamics exterior to the Mother Hedgehog at every scale, which Dudko and I established jointly.

Theorem G ([DLim26]). *Iterated preimages of the pseudo-Siegel disk of f_θ are uniformly quasiconformal and have uniformly controlled size.*

This theorem — proved, to our surprise, in the similar near-degenerate regime as Theorem A — is the geometric input that leads to the stronger uniform *butterfly bounds*. In joint work with Dudko and Lyubich, we applied these butterfly bounds to establish combinatorial rigidity for the full renormalization attractor.

Theorem H ([DLL26]). *Any two combinatorially equivalent bi-infinite neutral renormalization towers are conformally conjugate near their associated invariant Mother Hedgehogs.*

An alternative formulation is in terms of *neutral cascades*, global transcendental rescaled limits of f_θ near its critical value for any irrational θ . We proved that any two combinatorially equivalent neutral cascades are affinely conjugate. This theorem yields a complete dynamical universality picture for neutral maps, extending McMullen’s famous self-similarity result [McM98] depicted in Figure 3. After quotienting by conformal equivalence, we obtain the Full Renormalization Horseshoe, modeled by a shift map on its combinatorial data. The underlying combinatorial and arithmetic structure is worked out in a separated note of mine [Lim26c].

Our theorem unifies two previously well-developed regimes of neutral dynamics: bounded type [McM98, DLS20] and near-parabolic [IS06]. The former is closely related, via quasiconformal surgery, to the classical theory of critical circle maps and thus fit naturally into the broader real renormalization paradigm. The latter allows the chaotic Cremer phenomenon where the associated Mother Hedgehogs are not locally connected and which have no clear analog in real renormalization theory. In [DLL26], we have successfully developed a unified framework that accommodates such wild objects while retaining the rigidity and universality principles characteristic of renormalization.

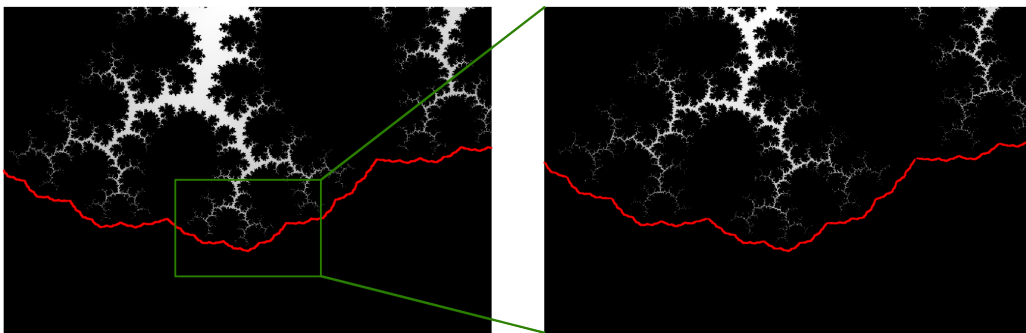


FIGURE 3. Self-similarity of the dynamical plane of f_θ near its critical value when θ is the golden mean irrational. The postcritical set is colored red.

3. ONGOING AND FUTURE PROJECTS

The results I mentioned above lead to several natural directions for future research. Here are the three main projects I am currently working on.

(1) **Uniform hyperbolicity of neutral renormalization**

With Dzmitry Dudko and Mikhail Lyubich, I am currently working to prove uniform hyperbolicity of the full neutral renormalization horseshoe, with exactly one unstable direction. Theorem H establishes the combinatorial side of the theory; uniform hyperbolicity would supply the corresponding analytic structure and is expected to have consequences on the fine geometric and measure-theoretic properties of the boundary of the Mandelbrot set.

(2) **Local connectivity of the Mandelbrot set**

A major long-term direction is the MLC. The remaining challenge is to establish combinatorial rigidity of quadratic polynomials that are infinitely renormalizable with unbounded combinatorics. With Jeremy Kahn, I am currently studying primitive near-neutral combinatorics, where we aim to establish the required *a priori bounds* by combining techniques from the near-degenerate regime [KL05, Kah06, DL22] with the regularization methods of [DL22]. The methods we are developing are expected to be applicable in other settings.

(3) **Topology of Mother Hedgehogs**

Another direction concerns the topology of the postcritical set of f_θ . Cheraghi [Che23, Che25] constructed a straight toy model predicting a trichotomy of possible topological types under suitable arithmetic conditions: a Jordan curve, a one-sided hairy Jordan curve, or a Cantor bouquet. I am currently working toward extending this picture to all irrational rotation numbers and all possible parabolic limits using pseudo-Siegel bounds, the estimates from [Lim26d], and a variant of Cheraghi's toy model. This will resolve the longstanding conjecture that every quadratic Siegel disk is a Jordan domain.

Lastly, below is a list of three additional directions I aim to pursue in the future. These projects should be tractable for strong graduate students.

(4) **Pseudo-Herman rings**

I plan to adapt the near-degenerate methods in [DL22, Lim25] to construct uniform *pseudo-Herman rings* — an annular analog of pseudo-Siegel disks — for cubic Blaschke products

$$g_{t,a}(z) = e^{2\pi it} z^2 \frac{z-a}{1-az}, \quad a > 3, t \in \mathbb{R},$$

across all irrational rotation numbers. This would yield a doubly-connected analog of Mother Hedgehogs, give a concrete example of Cremer phenomena in real dynamics for the first time, and shed light on the regularity of irrational Arnol'd tongues.

(5) **Biaccessibility of Cremer Julia sets**

Another longstanding open problem in neutral dynamics asks whether the neutral fixed point 0 is accessible from the exterior of the Julia set of f_θ when θ is non-Brjuno [McM94]. I expect that the butterfly bounds developed in the proof of Theorem H can resolve this, which would be a major step toward understanding the topology of the Julia set of f_θ , as outlined in [BO06].

(6) **Rigidity of multicritical quasicycle maps**

Lastly, I plan on adapting the near-degenerate methods of [DLim26] to develop the necessary butterfly bounds to establish: (a) rigidity of bounded type multicritical quasicycle maps, extending Theorem D; (b) rigidity of multicritical circle maps with arbitrary irrational rotation number, extending the work of Gorbovickis and Yampolsky [GY25] for bounded combinatorics.

REFERENCES

- [BO06] A. Blokh and L. Oversteegen. The Julia sets of quadratic Cremer polynomials. *Topol. Its Appl.*, 153:3038–3050, 2006.

- [Che13] D. Cheraghi. Typical orbits of quadratic polynomials with a neutral fixed point: Brjuno type. *Commun. Math. Phys.*, 332(3):999–1035, 2013.
- [Che19] D. Cheraghi. Typical orbits of quadratic polynomials with a neutral fixed point: Non-Brjuno type. *Ann. Sci. Éc. Norm. Sup.*, 52(1):59–138, 2019.
- [Che23] D. Cheraghi. Arithmetic geometric model for the renormalisation of irrationally indifferent attractors. *Non-linearity*, 36(12):6403–6475, 2023.
- [Che25] D. Cheraghi. Topology of irrationally indifferent attractors. *Ann. Sci. Éc. Norm. Sup.*, 6(58):1321–1383, 2025.
- [dF99] E. de Faria. Asymptotic rigidity of scaling ratios for critical circle mappings. *Ergod. Theory Dyn. Syst.*, 19(4):995–1035, 1999.
- [dFdM99] E. de Faria and Welington de Melo. Rigidity of critical circle mappings II. *J. Amer. Math. Soc.*, 13:343–370, 1999.
- [DL22] D. Dudko and M. Lyubich. Uniform a priori bounds for neutral renormalization, 2022. arXiv:2210.09280.
- [DL26a] D. Dudko and M. Lyubich. Uniform a priori bounds for neutral renormalization. Variation I: Sector renormalization, 2026. Manuscript.
- [DLim26] D. Dudko and W. R. Lim. Uniform bounds for bubbles of neutral quadratic polynomials, 2026. Manuscript.
- [DLL26] D. Dudko, W. R. Lim, and Mikhail Lyubich. Rigidity of the attractor of neutral renormalization, 2026. Manuscript.
- [DLS20] D. Dudko, M. Lyubich, and N. Selinger. Pacman renormalization and self-similarity of the Mandelbrot set near Siegel parameters. *J. Amer. Math. Soc.*, 33(3):653–733, 2020.
- [Dou87] A. Douady. Disques de Siegel et anneaux de Herman. In *Séminaire Bourbaki : volume 1986/87, exposés 669-685*, number 152-153 in Astérisque, pages 4, 151–172. Société mathématique de France, 1987.
- [DY25] K. Drach and J. Yang. Rigidity of bounded-type siegel polynomials, 2025. arXiv:2511.21246.
- [Ere20] A. Eremenko. Open problems session at the conference On Geometric Complexity of Julia Sets II. <https://www.impan.pl/konferencje/bcc/2020/20-juliasets2/problem-session.pdf>, 2020.
- [GY25] I. Gorbovickis and M. Yampolsky. Rigidity of analytic and smooth bi-cubic multicritical circle maps with bounded type rotation numbers. *Adv. Math.*, 481(110541), 2025.
- [IS06] H. Inou and M. Shishikura. The renormalization for parabolic fixed points and their perturbation, 2006. Preprint.
- [Kah06] J. Kahn. A priori bounds for some infinitely renormalizable quadratics: I. bounded primitive combinatorics, 2006. arXiv:math/0609045.
- [KL05] J. Kahn and M. Lyubich. The quasi-additivity law in conformal geometry. *Ann. Math.*, 169(2):561–593, 2005.
- [Lan88] O. E. Lanford. *Renormalization Group Methods for Circle Mappings*, pages 25–36. Springer US, 1988.
- [Lim25] W. R. Lim. A priori bounds and degeneration of Herman rings with bounded type rotation number. *Invent. Math.*, 242:827–893, 2025.
- [Lim26a] W. R. Lim. Rigidity of J-rotational rational maps and critical quasicycle maps. *Trans. Amer. Math. Soc.*, 379:2507–2567, 2026.
- [Lim26b] W. R. Lim. Hyperbolicity of renormalization of critical quasicycle maps. *Commun. Math. Phys.*, 407(88), 2026.
- [Lim26c] W. R. Lim. The combinatorics of sector renormalization, 2026. arXiv:2607.11408.
- [Lim26d] W. R. Lim. Lebesgue measure of the postcritical set of neutral quadratic polynomials, 2026. Manuscript.
- [McM94] C. McMullen. *Complex Dynamics and Renormalization*. Princeton University Press, 1994.
- [McM98] C. McMullen. Self-similarity of Siegel disks and Hausdorff dimension of Julia sets. *Acta Math.*, 180(2):247–292, 1998.
- [McM00] C. McMullen. Hausdorff dimension and conformal dynamics II: Geometrically finite rational maps. *Comment. Math. Helv.*, 75(4):535–593, 2000.
- [MS98] C. McMullen and D. Sullivan. Quasiconformal homeomorphisms and dynamics III. The Teichmüller space of a holomorphic dynamical system. *Adv. Math.*, 135(2):351–395, 1998.
- [PZ04] C. L. Petersen and S. Zakeri. On the Julia set of a typical quadratic polynomial with a Siegel disk. *Ann. Math.*, 159:1–52, 2004.
- [Rem09] L. Rempe. Rigidity of escaping dynamics for transcendental entire functions. *Acta Math.*, 203(2):235–267, 2009.
- [Sul83] D. Sullivan. Conformal dynamical systems. In *Geometric Dynamics*, volume 1007, pages 625–752. Springer, 1983.
- [Yam01] M. Yampolsky. The attractor of renormalization and rigidity of towers of critical circle maps. *Commun. Math. Phys.*, 218(3):537–568, 2001.
- [Yam02] M. Yampolsky. Hyperbolicity of renormalization of critical circle maps. *Publ. Math. Inst. Hautes Études Sci.*, 96:1–41, 2002.
- [Yam03] M. Yampolsky. Renormalization horseshoe for critical circle maps. *Commun. Math. Phys.*, 240(1–2):75–96, 2003.
- [Yoc95] J.-C. Yoccoz. Théorème de Siegel, nombres de Bruno et polynômes quadratiques. In *Petits diviseurs en dimension 1*, number 231 in Astérisque, pages 1–88. S. M. F., 1995.